

EXT I 2006 TERM 1

QUESTION ONE - (Start a new page)

marks

- (a) Differentiate $y = \ln(\cos^2 x)$ 2
- (b) The sides of a cube are decreasing at a constant rate of 2.5 cm s^{-1} . Find the rate at which the volume of the cube is changing when the sides are 15 cm. 2
- (c) Show that $k(4k+1)^{-1} + (4k+1)^{-1}(4k+5)^{-1} = (k+1)(4k+5)^{-1}$ 2
- (d) Find the exact sum of the first twenty terms of the series:
 $\log_a 4 + \log_a 16 + \log_a 64 + \dots$ 3

QUESTION TWO - (Start a new page)

- (a) Solve: (i) $\cos^2 x - \sin 2x = 0$ for $0 \leq x \leq 2\pi$ 3
- (ii) $1 = 2\log_{10} x - \log_{10}\left(\frac{x}{10} + 24\right)$ 3
- (b) Differentiate $3xe^x$ with respect to x , and hence or otherwise evaluate $\int_0^2 xe^x dx$. 3

QUESTION THREE - (Start a new page)

- (a) Water is pouring into a cone shaped funnel at a constant rate of $36 \text{ cm}^3 \text{ s}^{-1}$. If the diameter of the funnel is $\frac{3}{4}$ of its height, find the rate at which the depth of water is increasing when the height is 12 cm. Give your answer correct to three sig. figures. 3
- (b) A dinghy is being pulled towards a wharf at a constant rate of 15 m per minute. The rope is tied to the dinghy and the dinghy is 5 m below the wharf. Find the rate at which the:
- (i) rope is being drawn in when the dinghy is 12 m from the wharf. 3
- (ii) the angle between the rope and wharf is changing when the dinghy is 12 m from the wharf. 3

QUESTION FOUR - (Start a new page)

- (a) If $n! > 2^n$ for all integer values of n greater than 3, prove that $(n + 1)! > 2^{n+1}$ **3**
- (b) Given that $\int_0^k \frac{3x^2}{x^3 + 3} dx = \ln 10$, find the exact value of k . **2**
- (c) Prove by Mathematical Induction, that $n^3 + 2n$ is divisible by 3, for all positive integers n . **4**

QUESTION FIVE - (Start a new page)

Dwayne borrows \$200 000 which is to be repaid in equal monthly repayments of \$ x over 20 years. If interest is charged at 6% p.a. calculated monthly on the balance outstanding, find:

- (a) The amount owing after the first repayment. **1**
- (b) The amount of each monthly repayment to the nearest dollar. **3**
- (c) How long it would take to repay the same loan if Dwayne pays an extra \$100 every month from the very start? **3**
- (d) Assuming Dwayne makes the extra repayments of \$100, and after 5 years he wins \$50 000, can he pay out the balance of the loan? If not, how much more does he owe? **2**

QUESTION SIX - (Start a new page)

- (a) Express $0.\dot{5}\dot{0}$ as a geometric series and hence convert $0.\dot{5}\dot{0}$ to a rational number in its simplest form. **2**
- (b) Use Simpson's Rule with 5 functional values, to find the approximate area under the curve $y = \sin(e^{2x})$, the x -axis and the lines $x = 1$ to $x = 3$. Give your answer correct to two decimal places. **4**

QUESTION SIX - continued

- (c) Find the exact volume of the solid of revolution when the area under the curve $y = \cos 3x$, from $x = 0$ to $x = \frac{\pi}{6}$ is rotated about the x -axis. **3**

QUESTION SEVEN - (Start a new page)

- (a) Given that the sum of the infinite geometric series $1 + 2^n + 2^{2n} + \dots$ is 2. Find the exact value of n . **2**
- (b) Find $\int_0^1 e^{\ln 4x} dx$. **2**
- (c) MON is a quadrant of a circle centre O and radius 20cm. P is a point on the arc MN rotating about O at a constant rate, moving from M to N in 15 minutes. A is the total area of $\triangle OMP$ and $\triangle ONP$ in cm^2 .
- (i) Show that $A = 200(\sin\theta + \cos\theta)$, where θ is the angle MOP . **1**
- (ii) Find the exact rate at which A is changing when $\theta = \frac{\pi}{6}$. **4**

END OF PAPER

Question One

(a) $y = \ln(\cos^2 x)$

$$\frac{dy}{dx} = \frac{-2\cos x \sin x}{\cos^2 x} \quad (1)$$

$$= -2\tan x \quad (1)$$

(b) $\frac{dx}{dt} = -2.5$

$$V = x^3$$

$$\frac{dV}{dx} = 3x^2$$

$$\frac{dV}{dt} = ?$$

$$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt} \quad (1)$$

$$= 3x^2 \times -2.5$$

$$= -7.5x^2$$

When $x = 15\text{cm}$, $\frac{dV}{dt} = -1687.5 \quad (1)$

$\therefore$ Volume is decreasing at $1687.5\text{cm}^3/\text{s}$

(c) $\frac{k}{(4k+1)} + \frac{1}{(4k+1)(4k+5)} = \frac{k+1}{(4k+5)}$

$$\text{LHS} = \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$$

$$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)} \quad (1)$$

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}$$

$$= \frac{k+1}{(4k+5)} \quad (1)$$

$$= \text{RHS}$$

(d) $a = \log_4 4 = 2\log_2 2 \quad (1)$

$$2\log_2 2 + 4\log_2 2 + 6\log_2 2 + \dots$$

$$S_{20} = \frac{20}{2} \{ 4\log_2 2 + 19(2\log_2 2) \} \quad (1)$$

$$= 420\log_2 2 \quad (1)$$

Question Two

(a) i) $\cos^2 x - \sin 2x = 0$

$$\cos^2 x - 2\sin x \cos x = 0$$

$$\cos x (\cos x - 2\sin x) = 0 \quad (1)$$

$$\cos x = 0 \quad \text{or} \quad \tan x = \frac{1}{2} \quad (1)$$

$$(1) \therefore x = \frac{\pi}{2}, \frac{3\pi}{2} \quad \text{or} \quad x = 0.4636, 3.6052$$

ii) $1 = 2\log_{10} x - \log_{10} \left(\frac{x}{10} + 24 \right)$

$$1 = \log_{10} x^2 - \log_{10} \left(\frac{x}{10} + 24 \right)$$

$$1 = \log_{10} \left(\frac{x^2}{\frac{x}{10} + 24} \right)$$

$$10 = \frac{x^2}{\frac{x}{10} + 24} \quad (1)$$

$$x^2 = x + 240$$

$$0 = x^2 - x - 240$$

$$0 = (x-16)(x+15) \quad (1)$$

$$\therefore x = 16 \quad \text{or} \quad -15$$

but $x > 0$ as you can't take a log of a negative number $\therefore x = 16$ only (1)

(b) $y = 3xe^x$

$$\frac{dy}{dx} = 3xe^x + 3e^x \quad (1)$$

$$\int_0^2 \frac{d(3xe^x)}{dx} dx = \int_0^2 3xe^x dx + \int_0^2 3e^x dx$$

$$\frac{1}{3} [3xe^x]_0^2 = \int_0^2 3xe^x dx + [e^x]_0^2 \quad (1)$$

$$2e^2 - 0 = \int_0^2 3xe^x dx + e^2 - 1$$

$$\therefore \int_0^2 3xe^x dx = e^2 + 1 \quad (1)$$

Question 3

a) $\frac{dV}{dt} = 36 \text{ cm}^3/\text{s}$



$$2r = \frac{3}{4}h$$

$$r = \frac{3}{8}h \quad (1)$$

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{\pi}{3} \cdot \frac{9h^3}{64}$$

$$V = \frac{3\pi h^3}{64}$$

$$(1) \quad \frac{dV}{dh} = \frac{9\pi h^2}{64}$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

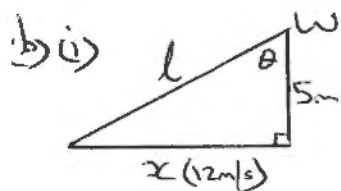
$$= \frac{64}{9\pi h^2} \times 36$$

$$= \frac{256}{\pi h^2} \quad (1)$$

when $h=12$, $\frac{dh}{dt} = 0.565884242$

∴ Height's increasing at 0.566 cm/s

(* This question is not the target question for rounding off errors!)



$$L = \sqrt{25 + x^2} \quad (\text{Pythagoras})$$

$$\frac{dL}{dx} = \frac{x}{\sqrt{25 + x^2}} \quad (1)$$

$$\frac{dx}{dt} = 15 \text{ m/min}$$

$$\frac{dL}{dt} = ?$$

$$\frac{dL}{dt} = \frac{dL}{dx} \times \frac{dx}{dt}$$

$$= \frac{x}{\sqrt{25 + x^2}} \times 15 \quad (1)$$

when $x=12$, $\frac{dL}{dt} = 13 \frac{1}{13}$

∴ rope is being drawn in at $13 \frac{1}{13} \text{ m/min}$

ii) $\tan \theta = \frac{x}{5}$

$$x = 5 \tan \theta$$

$$\cos \theta = \frac{5}{13}$$

$$\frac{dx}{d\theta} = 5 \sec^2 \theta \quad (1)$$

$$\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt}$$

$$= \frac{1}{5} \cos^2 \theta \times 15$$

$$(1) = \frac{1}{5} \times \frac{25}{169} \times 15$$

$$= \frac{-75}{169}$$

∴ Angle's decreasing at $\frac{75}{169} \text{ rad/min}$

Question 4

(a) given $n! > 2^n$ for $n \geq 4$

prove $(n+1)! > 2^{n+1}$

$$\text{now } (n+1)! = (n+1)(n!) \quad (1)$$

$$\text{so } (n+1)n! > (n+1)2^n \quad (\text{as } n! > 2^n)$$

$$\text{and } (n+1)2^n > 2(2^n) \quad \text{as } n \geq 4 \quad (1)$$

$$\text{so, } (n+1)n! > (n+1)2^n > 2^{n+1}$$

$$\therefore (n+1)n! > 2^{n+1} \quad (1)$$

$$\text{i.e. } (n+1)! > 2^{n+1}$$

b) $\int_0^K \frac{3x^2}{x^3+3} dx = \ln 10$

$$\ln 10 = [\ln(x^3+3)]_0^K \quad (1)$$

$$\ln 10 = \ln(K^3+3) - \ln 3$$

$$\ln 30 = \ln(K^3+3)$$

$$K^3+3=30$$

$$K^3=27$$

$$K=3 \quad (1)$$

c) step 1: Prove true for $n=1$,

$$(1+2)=3$$

∴ which is divisible by 3 ∴ true for $n=1$ (1)

step 2: Assume true for $n=k$,

$$\text{i.e. } k^3+2k \text{ is divisible by 3} \quad (1)$$

$$\therefore k^3+2k=3M \quad (\text{where } M \text{ is integer})$$

step 3: Prove true for $n=k+1$,

i.e. show that $(k+1)^3+2(k+1)$ is divisible by 3

$$\text{LHS} = k^3+3k^2+3k+1+2k+2 \quad (1)$$

$$= k^3+2k+3k^2+3k+3$$

$$= 3M+3(k^2+k+1) \quad (\text{by assumption})$$

$$= 3(M+k^2+k+1) \quad (1)$$

which is divisible by 3

step 4: Since true for $n=1$, ~ ~ ~

Question 5

(a) let A_n = amount owing after n^{th} repayment

$$A_1 = 200000(1 + \frac{1}{200}) - P$$

$$A_1 = 200000(1 + \frac{1}{200}) - x \quad (1)$$

$$(b) A_2 = 200000(1 + \frac{1}{200})^2 - x[(1 + \frac{1}{200}) + 1]$$

$$\vdots$$

$$A_{240} = 200000(1 + \frac{1}{200})^{240} - x[(1 + \frac{1}{200})^{239} + \dots + 1]$$

$$\text{but } A_{240} = 0$$

$$(1) \therefore x = \frac{200000(1 + \frac{1}{200})^{240}}{\frac{1}{200} + \frac{1}{200} + \dots + 1}$$

denom. is a G.P., $a=1, r=1 + \frac{1}{200}, n=240$

$$(1) S_{240} = 1 \frac{(1 + \frac{1}{200})^{240} - 1}{\frac{1}{200}} = 462.0409$$

$$\therefore x = \frac{200000(1 + \frac{1}{200})^{240}}{462.0409}$$

$$(1) = \$1432.86$$

$$\therefore x = \$1433 \text{ (nearest dollar)}$$

$$\hookrightarrow x = 1433 + 100 = \$1533$$

$$n = ?$$

$$(1) 1532 = \frac{200000(1 + \frac{1}{200})^n}{\frac{(1 + \frac{1}{200})^n - 1}{\frac{1}{200}}}$$

$$1.532 = \frac{(1 + \frac{1}{200})^n}{(1 + \frac{1}{200})^n - 1}$$

$$1.532 (1 + \frac{1}{200})^n - 1.532 = (1 + \frac{1}{200})^n$$

$$0.532 (1 + \frac{1}{200})^n = 1.532$$

$$(1) \frac{1}{200}^n = 2.87969248$$

$$\log_{1/200} 2.87969 = n$$

$$(1) n = \frac{\log 2.87969}{\log_{10} 1/200}$$

$$n = 212.06493$$

$\therefore$ It takes 212 months.

$$d) A_{60} = 200000(1 + \frac{1}{200})^{60} - 1533(1 + \frac{1}{200})^{59} + \dots + 1$$

$$= 200000(1.005)^{60} - 1533(\frac{1 + \frac{1}{200} - 1}{\frac{1}{200}})$$

$$(1) = \$162812.57$$

$$A_{60} - \$50000 = \$112812.57$$

$\therefore$ Dwayne can't pay off the loan
(1) as he still owes \$112812.57.

Question 6

$$a) 0.\dot{5}\dot{0} = 0.50 + 0.0050 + 0.00005 \quad \textcircled{1}$$

$$r = \frac{1}{100} \quad a = \frac{1}{2}$$

$$S_{\infty} = \frac{\frac{1}{2}}{1 - \frac{1}{100}}$$

$$= \frac{50}{99} \quad \textcircled{1}$$

$$b) A = \int_1^3 \sin(e^{2x}) dx$$

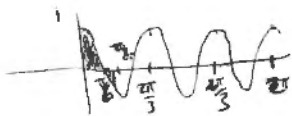
$$\textcircled{1} A = \frac{2-1}{6} [f(1) + 4f(1\frac{1}{2}) + 2f(2) + 4f(2\frac{1}{2}) + f(3)]$$
$$= \frac{1}{6} [\sin(e^2) + 4\sin(e^3) + 2\sin(e^4) + 4\sin(e^5) + \sin(e^6)]$$

$$\textcircled{1} = \frac{1}{6} [\sin(7.38906) + 4\sin(20.0855) + 2\sin(54.598) + 4\sin(148.4132) + \sin(403.4288)]$$
$$= 1.028251124 \quad \textcircled{1}$$

$$* = 1.03 \text{ (2 dec. places)} \quad \textcircled{1}$$

(* This is the target question if they have answered correct to 2 dec. places, deduct a mark.).

$$c) V = \pi \int_0^{\pi/6} \cos^2 3x dx$$



$$= \frac{\pi}{2} \int_0^{\pi/6} (\cos 6x + 1) dx \quad \textcircled{1}$$

$$= \frac{\pi}{2} \left[\frac{1}{6} \sin 6x + x \right]_0^{\pi/6} \quad \textcircled{1}$$

$$= \frac{\pi}{2} \left[\left(\frac{1}{6} \sin \pi + \frac{\pi}{6} \right) - (0 + 0) \right]$$

$$= \frac{\pi^2}{12} \text{ units}^3 \quad \textcircled{1}$$

QUESTION SEVEN

$$a) 1 + 2^n + 2^{2n} + \dots = 2$$

$$a=1 \quad r=2^n$$

$$S_{\infty} = \frac{a}{1-r}$$

$$2 = \frac{1}{1-2^n} \quad (1)$$

$$2(1-2^n) = 1$$

$$2 - 2^{n+1} = 1$$

$$2^{n+1} = 1$$

$$\therefore n+1 = 0$$

$$\therefore n = -1 \quad (1)$$

$$b) \int_0^1 e^{\ln 4x} dx = \int_0^1 4x dx \quad (1)$$

$$= [2x^2]_0^1$$

$$= 2 - 0$$

$$= 2 \quad (1)$$

$$c) i) A = \frac{1}{2}r^2 \sin \theta + \frac{1}{2}r^2 \sin(90-\theta)$$

$$= \frac{1}{2}(20)^2 \sin \theta + \frac{1}{2}(20)^2 \cos \theta \quad (1)$$

$$= 200(\sin \theta + \cos \theta)$$

$$ii) \frac{dA}{d\theta} = 200(\cos \theta - \sin \theta)$$

$$\frac{dl}{dt} = \frac{1}{4} \left(\frac{2\pi 20}{15} \right) / \text{min}$$

$$= \frac{10\pi}{15}$$

$$= \frac{2\pi}{3} \text{ cm/min} \quad (1)$$

$$l = 20\theta \quad (l=r\theta)$$

$$\frac{dl}{d\theta} = 20$$

$$\frac{d\theta}{dt} = \frac{d\theta}{dl} \times \frac{dl}{dt}$$

$$= \frac{1}{20} \times \frac{2\pi}{3}$$

$$= \frac{\pi}{30} \quad (1)$$

$$\frac{dA}{dt} = \frac{dA}{d\theta} \times \frac{d\theta}{dt}$$

$$= 200(\cos \theta - \sin \theta) \times \frac{\pi}{30}$$

$$= \frac{20\pi}{3} (\cos \theta - \sin \theta) \quad (1)$$

$$\text{when } \theta = \pi/6, \frac{dA}{dt} = \frac{20\pi}{3} (\cos \frac{\pi}{6} - \sin \frac{\pi}{6})$$

$$= \frac{20\pi}{3} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right)$$

$$= \frac{10\pi}{3} (\sqrt{3} - 1) \quad (1)$$

$\therefore$ Area is increasing at

$$\frac{10\pi}{3} (\sqrt{3} - 1) \text{ cm}^2/\text{s}.$$

End of Paper!!